INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)

CIP-B101: Basic Computer Skills for Digital Forensics

Lab 1B: PC System Fundamentals, Disk Sectors and Partition Analysis

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1. LAB OVERVIEW & OBJECTIVES

This practical laboratory focuses on establishing fundamental computer system competency for digital forensic investigations. Forensic examiners must safely inspect system hardware, evaluate memory structures, inspect storage devices, and analyze partition layouts without altering digital evidence. All tasks executed during this session strictly adhered to read-only forensic observation practices.

2. TOOLS & COMMANDS REFERENCE

Tool / Utility	Purpose in Investigation	Execution Command
lsblk	Inspect block devices, partitions, sizes, and mount points.	`lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINT`
fdisk	Read-only partition table sector layout and sector boundaries.	`sudo fdisk -l`
blkid	Locate block device attributes, file system UUIDs, and volume names.	`sudo blkid`
df	Report file system disk space usage and mount point status.	`df -h`
ps	Snapshot running processes, Process IDs (PIDs), and resource use.	`ps aux | head -n 15`

3. EVIDENCE LAYER MAPPING

System Layer	Example Evidence	Storage Location	Forensic Relevance
User Layer	Opened documents, web history, typed commands	Home directory, `.bash_history`, browser caches	Directly attributes actions and intent to a specific user account.
Application Layer	App configs, database logs, chat history	`/etc/`, `/var/log/`, `~/.config/`	Identifies third-party software usage and user interactions.
OS Layer	System logs, service states, user accounts	`/var/log/syslog`, `/etc/passwd`	Establishes system boot times, logins, and service execution.
Disk Layer	Raw partitions, unallocated space, file systems	Block devices (`/dev/sda1`), raw sector space	Primary target for data recovery, metadata extraction, and timelines.
Memory Layer	Active processes, volatile RAM, network sockets	Process table, `/proc` file system, RAM dump	Captures live state, unencrypted credentials, and malicious code.

4. PROGRAM, PROCESS, AND THREAD INVESTIGATION

During live system triage, digital forensic examiners must distinguish between static binaries and active memory execution artifacts:

- **Program:** A passive file stored on secondary storage (disk) containing compiled executable code (e.g., `/sbin/init`).
- **Process:** An active instance of a program loaded into volatile memory (RAM) with allocated memory space, file descriptors, and a unique Process ID (PID).
- **Thread:** The smallest unit of execution managed by the operating system kernel within a process. Multiple threads share the parent process's memory space.

Recorded Terminal Process Snapshot

```
USER PID %CPU %MEM VSZ RSS TTY STAT START TIME COMMAND root 1 0.0 0.7 24012 14300 ? Ss 00:53 0:02 /sbin/init splash root 2 0.0 0.0 0 0 ? S 00:53 0:00 [kthreadd] root 3 0.0 0.0 0 0 ? I< 00:53 0:00 [pool_workqueue_release] root 15 0.0 0.0 0 0 ? I 00:53 0:00 [rcu_tasks_rude_kthread]
```

5. DISK SECTOR & PARTITION ANALYSIS

Read-only inspection of the virtual hard drive (`/dev/sda`) was conducted using `fdisk -l` and `lsblk`. The system utilizes a 512-byte sector architecture.

```
Disk /dev/sda: 80.09 GiB, 86000000000 bytes, 167968750 sectors Disk model: VBOX HARDDISK Units: sectors of 1 * 512 = 512 bytes Sector size (logical/physical): 512 bytes / 512 bytes Disklabel type: dos (MBR) Device Boot Start End Sectors Size Id Type /dev/sda1 * 2048 167968749 167966702 80.1G 83 Linux
```

Part G Step-by-Step Size Calculation

1. Partition Sector Count Calculation:

Total Sectors = End Sector - Start Sector + 1

Total Sectors = 167,968,749 - 2,048 + 1 = 167,966,702 sectors

2. Raw Byte Storage Calculation:

Total Bytes = 167,966,702 × 512 bytes = 85,998,951,424 bytes

3. Conversion to Binary Gigabytes (GiB):

Size in GiB = 85,998,951,424 / (1024 × 1024 × 1024) = 85,998,951,424 / 1,073,741,824 ≈ 80.09 GB

6. PART G: PRACTICAL ASSIGNMENT WORKSHEETS

Task Required	Terminal / Method Reference	Findings & Calculations
Identify Partition 1	`sudo fdisk -l` output	`/dev/sda1` (Bootable Linux Primary Partition)
Sector Boundaries	Sector `Start` and `End` values	Start: `2048` | End: `167968749` | Sectors: `167966702`
Calculate Size (GB)	Manual formula calculation	**80.09 GB** (Calculated raw sector size)
Identify Partition 2	Disk partition table inspection	*N/A* (Virtual machine uses a single main root partition)
OS Size Comparison	`lsblk` vs `df -h`	`lsblk` reports **80.1G**; `df -h` reports **79G** total mounted space.
Explain Difference	File system overhead analysis	Discrepancies arise from `ext4` metadata, journaling space, and decimal (GB) vs binary (GiB) units.

7. PC BOOT PROCESS & PARTITIONING STANDARDS

The Standard PC Boot Sequence

1. **Power-On and POST:** The motherboard firmware performs hardware self-checks on CPU, memory, and primary buses.
2. **Firmware Initialization (BIOS/UEFI):** System settings are loaded from CMOS; firmware identifies bootable storage devices.
3. **Boot Record Execution:**
   - MBR System: Sector 0 is read into memory, executing bootstrap code to locate the active partition.
   - UEFI System: Reads EFI System Partition (ESP) to execute designated `.efi` boot loaders.
4. **Boot Loader Stage:** Boot managers like GRUB load kernel images and initial RAM disks (`initrd`) into memory.
5. **Kernel & OS Control:** Operating system kernel initializes drivers, mounts the root filesystem (`/`), and starts `systemd`.

Comparative Technical Analysis: MBR vs. GPT

Feature	MBR (Master Boot Record)	GPT (GUID Partition Table)
Age & Architecture	Legacy architecture introduced in 1983 (IBM PC DOS).	Modern standard associated with UEFI systems.
Disk Size Capacity	Maximum addressable size of 2.2 TB (32-bit LBA).	Supports up to 9.4 ZB (64-bit logical block addressing).
Partition Limits	Maximum 4 primary partitions (or 3 primary + 1 extended).	Supports up to 128 primary partitions natively.
Redundancy & Repair	Single point of failure (Sector 0 only). Corrupt MBR breaks boot.	Primary header at start; backup header at end of disk.

8. FILE SYSTEM IDENTIFICATION & FORENSIC VALUE

Command-line evaluation via `blkid` confirmed that target partition `/dev/sda1` utilizes the **ext4** (Fourth Extended File System) file system.

```
/dev/sda1: LABEL="root" UUID="af89d218-8d5b-4054-a5b7-db050b5f62a7" BLOCK_SIZE="4096"
TYPE="ext4" PARTUUID="06c77605-01
```

Forensic Significance of File System Structure:
File systems define how data bytes are organized into human-readable files. For digital forensic examiners, understanding `ext4` structures (such as *inodes*, *extent trees*, and *journals*) allows for metadata recovery, accurate timestamp reconstruction (atime, mtime, ctime, crtime), and deleted file carving from unallocated space blocks.

9. CONCLUSION

This practical laboratory successfully demonstrated safe, non-destructive evidence gathering procedures within a Linux forensic environment. System components, process states, block devices, sector sizes, and file systems were inspected using native read-only terminal tools. Manual mathematical validation confirmed that partition `/dev/sda1` spans `167,966,702` sectors, equaling `80.09 GB` of storage capacity. All findings have been logged to support reproducible forensic reporting.

> **Forensic Integrity Assurance:** All commands executed during this investigation were non-destructive read-only operations (`lsblk`, `fdisk -l`, `blkid`, `df -h`). No disk write operations, partitioning alterations, or file modifications were performed on the evidence source.